

Joan Vila Orús

R&D Engineer | Applied AI & Computer Graphics | C++ & Python Developer

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EDUCATION

Universitat Politècnica de Catalunya (UPC - FIB)	Barcelona, Spain
<i>Bachelor of Science in Computer Engineering (Computing Specialization)</i>	<i>Sep. 2022 – Expected July 2026</i>
• Focus: High-Performance Computing, Artificial Intelligence, Computer Graphics (OpenGL), Software Architecture. • Bachelor Thesis: "Real-Time Embodied AI Agent in VR". Low-latency integration of LLMs into 3D avatars.	
Sapienza Università di Roma	Rome, Italy
<i>Erasmus+ Exchange Program (Computer Engineering)</i>	<i>Feb. 2026 – July 2026</i>
• Focus Areas: Artificial Intelligence, Robotics & Computer Vision. • International mobility program. Coursework completed in English & Italian.	

TECHNICAL SKILLS

Core Languages: C++ (STL, Memory Management), **Python** (Advanced), C#, SQL, Bash.
AI & Data Science: PyTorch, Transformers (BERT), Google Vertex AI, LLMs (Mistral/Llama..), Pandas, OpenCV.
Graphics & XR: Unity 3D, Unreal Engine Basics, OpenGL, Shader Graph, VR Development (Oculus SDK).
Infrastructure & Tools: Linux/Unix, Docker, Git, Cloud Computing (GCP/AWS), LaTeX.
Languages: Spanish (Native), Catalan (Native), English (Professional/FCE).

RESEARCH & WORK EXPERIENCE

Machine Learning Engineer Intern (Research Unit)	Jan. 2025 - Present
<i>Hospital Clínic de Barcelona</i>	<i>Barcelona, Spain</i>
• Engineered privacy-preserving NLP pipelines to anonymize sensitive medical records using BERT-based models. • Deployed and validated models on Google Vertex AI, integrating synthetic data generation to improve model robustness. • Designed rigorous benchmarking protocols to evaluate model utility vs. privacy leakage with high autonomy. • Collaborated with clinical teams to translate complex medical requirements into scalable software solutions.	
Activity Coordinator & Resident Assistant	2024 - 2025
<i>RESA Lesseps</i>	<i>Barcelona, Spain</i>
• Managed logistics and crisis resolution for a ~300 student residence, fostering leadership in an international environment.	

KEY TECHNICAL PROJECTS

Real-Time Embodied AI Agent in XR C#, Unity, C++, Python, WebSocket	GitHub
• Architected a low-latency bi-directional bridge (~200ms) between a Unity VR frontend and a local Python inference backend (Ollama/Llama). • Implemented procedural lip-sync and animation blending in real-time driven by TTS audio waveforms. • Optimized inference performance for consumer hardware, managing asynchronous data streams.	
The Rogue Taxi: Adaptive Negotiation AI Python, LangChain, FAISS (VectorDB)	GitHub
• Developed a game engine where an AI antagonist dynamically adapts difficulty using RAG. • Implemented a self-improving memory system: the agent embeds successful player arguments to counter future strategies.	
Assistive Object Detection System Python, OpenCV, YOLOv8	Ongoing
• Built a real-time object detection pipeline for visually impaired navigation support. • Optimized YOLO inference for edge deployment, mapping visual inputs to spatial auditory feedback.	

INTERESTS

Research: Real-Time Rendering, Generative AI for 3D Content, Medical Imaging AI, HCI in VR.
Other: Basketball Coach (2 years), Open Source Contributor